Voice Agentic Smart Reader Application in Sinhala and Tamil for Visually Impaired People

25-26J-125

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Voice Agentic Smart Reader Application in Sinhala and Tamil for Visually Impaired People

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DATA SCIENCE

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Assistive camera guidance system with integrated OCR

Proven Gap:
Visually impaired users struggle to capture clear, aligned, and oriented document images, reducing OCR accuracy and reading clarity.

Solution

Creative Solution:
Assistive mobile system with real-time camera guidance (voice + haptic), image quality checks, orientation detection, and OCR.

Knowledge Gap / Problem Definition:

- Existing OCR and camera apps assume sighted users and lack real-time guidance for visually impaired individuals.
- Current systems are not optimized for Sinhala and Tamil scripts, leading to low OCR accuracy and poor accessibility.

Shortcoming of existing system	Proposed Solution
OCR tools are mainly optimized for English and a few global languages.	Develop OCR models adapted to Sinhala and Tamil scripts.
No real-time guidance (voice/haptic) to help visually impaired users align or focus the camera.	Introduce voice and haptic feedback for positioning, alignment, and focus during image capture.
Existing apps do not reliably detect page orientation, leading to rotated or inverted text.	Integrate orientation detection with automatic correction before OCR processing.
Poor image quality (blur, glare, cropping) lowers OCR accuracy and readability.	Implement image quality assurance (focus check, boundary detection, clarity feedback) before running OCR.

Assistive camera guidance system with integrated OCR

Key Domains of specialization:

1. Data Science & Information Management
   A. Dataset collection and annotation
   B. Data augmentation
   C. Ethical handling and privacy

2. Artificial Intelligence & Machine Learning
   A. Supervised learning
   B. Deep learning
   C. Natural Language Processing

3. Human-Computer Interaction (HCI) & Edge Computing
   A. Designing non-visual interaction methods
   B. Performance optimization
   C. On-device inference

Latest technologies utilize:

1. Computer Vision
   A. Object detection models
   B. Hough Transform + deep orientation classifiers

2. Machine Learning:
   A. Transformer-based OCR models
   B. CNN + Attention models
   C. Natural Language Processing
   D. Computer Vision

3. Mobile & Edge Computing
   A. TensorFlow Lite / Core ML
   B. Model Quantization & Pruning

4. Human-Computer Interaction (HCI)
   A. Android TalkBack
   B. Android vibration engines

Assistive camera guidance system with integrated OCR

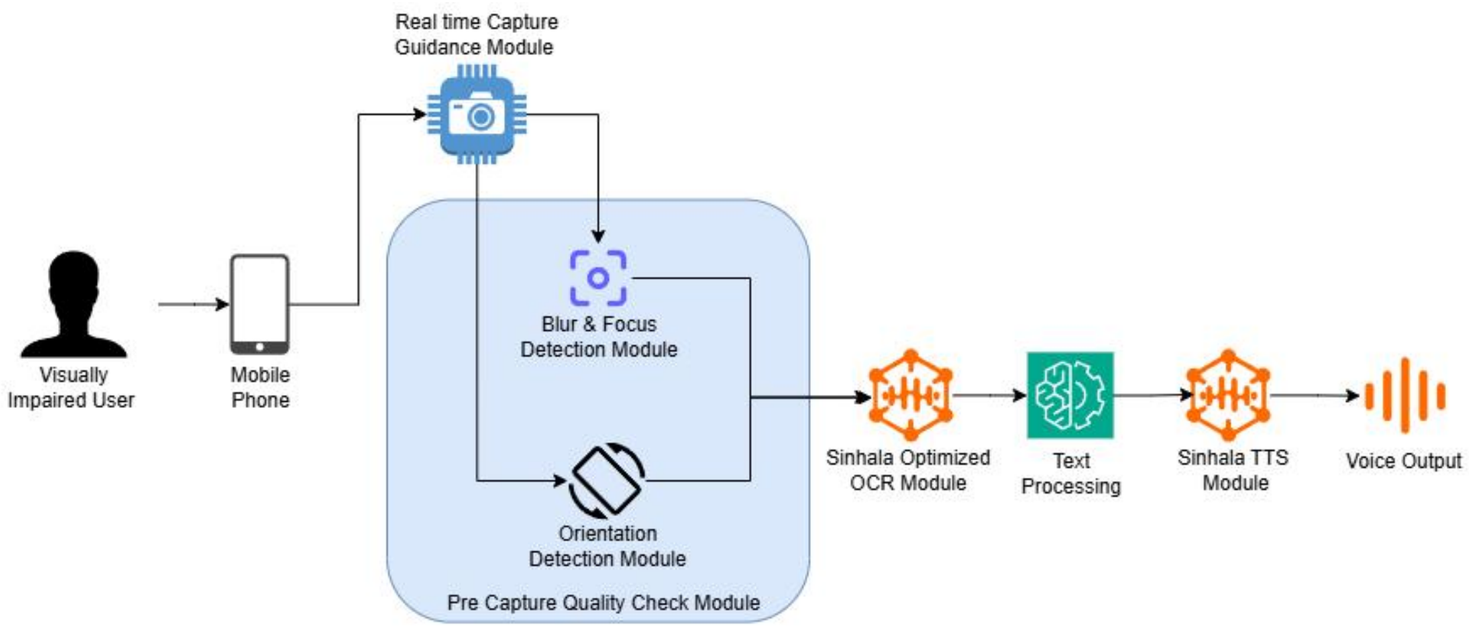
Evaluate/Prove Success:

- Capture success rate (readable page on first try)
- Comparison of Page corner/Intersection over Union (IoU)
- OCR (Sinhala/Tamil) Character Error Rate
- On-device latency (guidance loop)
- User Studies (Feedbacks)
- Usability Scores

Data availability & Ethical clearance:

- Public printed materials in Sinhala/Tamil (textbooks, newspapers) for non-personal pages.
- Small in-the-wild set captured with consent for real handheld variability (lighting, shadows).
- Immediate deletion of captured pages unless the user saves.
- secure storage with device-level encryption.

Assistive camera guidance system with integrated OCR



- The app greets the user with a voice prompt and is ready for document capture.
- The user places the document (e.g., a Sinhala or Tamil newspaper, book, or form) on a flat surface.
- Using the phone's camera, the app provides **real-time guidance**:
  - Voice instructions like *"Move phone left"*, *"Tilt slightly forward"*.
  - Haptic signals when the page is correctly aligned.
- Once the system detects clear alignment and focus, it gives a **short vibration + voice alert**
- The image is automatically captured (or the user can say "Capture").
- The app checks for blur, glare, and orientation.
- If issues are found, it alerts the user.
- If clear, the OCR extracts Sinhala/Tamil text.

Functional requirements	Non functional requirements
Camera Guidance – Provide real-time voice and haptic feedback	**Performance** – Guidance feedback latency < 3ms
Orientation Detection – Detect and auto-correct page orientation	**Accuracy** – OCR accuracy (Character Error Rate) ≤ 10%
Image Quality Assurance – Check for blur, glare, cropping	**Usability** – Simple, voice-driven interface for visually impaired users.
OCR Processing – Extract text from captured images with high accuracy.	**Security & Privacy** – All processing on-device by default.
Voice Commands – Allow hands-free control	**Ethical Compliance** – Must follow ethical data handling

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Data Science

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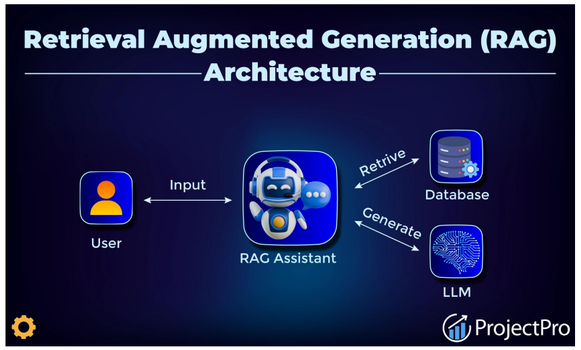
Intelligent Segmentation and RAG-based Summarization with Non-Linear Voice Navigation for Sinhala OCR Text

•**Proven Gap:** Existing Sinhala assistive tools are limited to basic TTS or extractive summaries, forcing linear reading and failing to provide personalized, non-linear navigation for visually impaired users.

•**Creative Solution:** Develop robust segmentation tailored for Sinhala, introduce RAG-based personalized summarization for low-resource languages, and implement an interactive, non-linear, voice-assisted navigation system enabling preference-driven exploration.

•**Knowledge Gap / Problem Definition:** Existing Sinhala assistive tools are limited to basic Text-to-Speech (TTS) or extractive summarization, which enforce linear reading and fail to support comprehension of long, complex documents.
Prior studies address isolated tasks (OCR, segmentation, or summarization) but lack an integrated accessibility-first solution for visually impaired users.

ASPECT	Shortcomings of Existing Systems	Proposed Solution
Language Coverage	High-resource languages (English); Sinhala/Tamil TTS	Full Sinhala support; non-linear voice navigation
Summarization	Summaries are often contextually shallow and not personalized.	RAG-based personalized summarization for low-resource languages.
Content Segmentation	Segmentation is weak, especially for noisy Sinhala documents.	Intelligent segmentation integrated with summarization workflow
Navigation	Linear TTS playback	Voice-assisted non- linear navigation; users can jump between sections and explore content based on preferences
Integration	Task-specific tools	Unified workflow combining segmentation, RAG-based summarization, and voice navigation

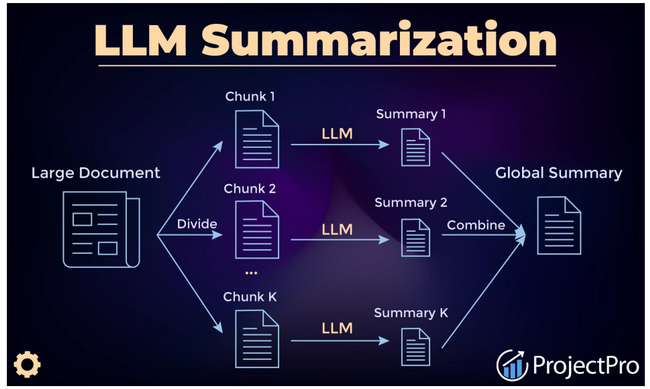


9/8/2025

Intelligent Segmentation and RAG-based Summarization with Non-Linear Voice Navigation for Sinhala OCR Text

Key Domains of Specialization

- NLP: Document segmentation, summarization, LLM fine-tuning
- Information Retrieval (IR): RAG pipelines for Sinhala
- Speech Technologies: STT, TTS and voice interaction
- HCI: accessible, non-linear, voice-assisted navigation



Latest Technologies Utilized

- Segmentation: Rule-based + multilingual transformer models (XLM-R, mBERT, IndicBERT)
- Summarization: OpenAI GPT-3 API for abstractive summarization.
- RAG framework: Fine-tuned Hugging Face Transformers (mT5, mBART, LLaMA)
- Voice Navigation: Google STT, Whisper, Sinhala TTS
- Vector Retrieval: FAISS / Pinecone with embeddings from multilingual LLMs.

Intelligent Segmentation and RAG-based Summarization with Non-Linear Voice Navigation for Sinhala OCR Text

Validation & Data/Ethics

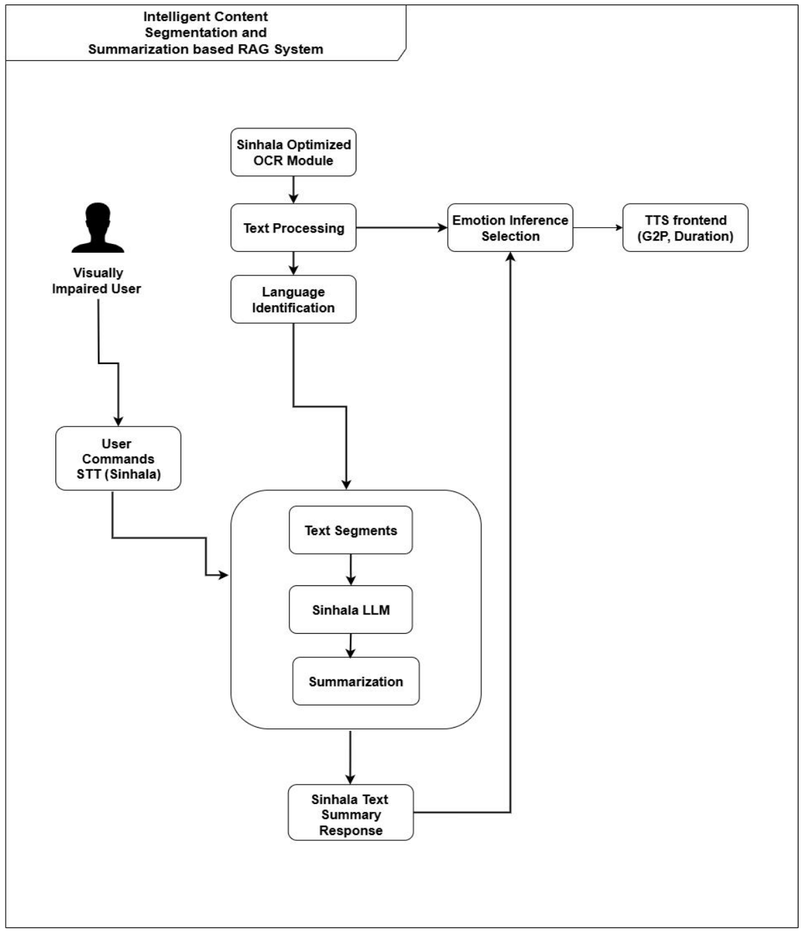
How to Evaluate/Prove Success

- Segmentation: Precision, Recall, F1-score, WindowDiff.
- Summarization (LLMs): ROUGE, BLEU, BERTScore + human evaluation(fluency, factual accuracy).
- Voice Navigation: Usability testing with visually impaired users(measure navigation time reduction, comprehension gain).
- Expected Accuracy:
    - Segmentation ≥ 85% (Pk/WindowDiff).
    - Summarization ≥ 20% improvement with LLMs + RAG vs. extractive baselines.

Data Availability & Ethical Clearance

- Data Sources: Sinhala corpora (news, academic, legal, literary), OCR datasets.
- LLM Training Data: Open-source Sinhala text collections; fine-tuned for domain tasks.
- Ethical Considerations:
    - Secure handling of text datasets.
    - Informed consent from visually impaired participants.
    - Accessibility-first design, minimizing bias in language representation.

Intelligent Segmentation and RAG-based Summarization with Non-Linear Voice Navigation for Sinhala OCR Text



- A visually impaired user takes a photo of a Sinhala document.
- OCR + segmentation converts it into structured text.
- User issues a voice query ("Give me a short summary of Section 2").
- System retrieves + summarizes via RAG-powered LLMs, then reads aloud through TTS.
- User navigates non-linearly: "Jump to conclusion," "Expand summary," etc.

Functional requirements	Non functional requirements
• OCR integration for Sinhala text.	• Fast summarization response (<3s latency)
• Accurate segmentation & summarization pipeline.	• High accuracy (≥20% better than extractive baselines)
• Voice command recognition and execution.	• Simple, voice-first interface
• TTS playback in Sinhala.	• User-friendly and accessible for visually impaired users
	• Scalable to handle long academic/legal documents
	• Modular and maintainable for future LLM/TTS upgrades

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Data Science

On-Device Tamil LLM

Proven Gap: Tamil NLP systems lack accurate summarization, topic extraction, and Q&A in low-resource, offline settings. Current solutions are cloud-based, heavy, and inaccessible to visually impaired users.

Solution

- **Creative Solution:** Develop a **voice-driven, on-device Tamil LLM Smart Reader** that provides real-time summarization, topic extraction, and interactive Q&A. Optimized for mobile devices, it ensures offline access, privacy, and accessibility

Knowledge Gap / Problem Definition: Tamil lacks curated summarization/Q&A datasets, tailored LLMs, and interactive assistive applications. Current systems depend on cloud resources, making them unsuitable for low-connectivity areas and inaccessible for visually impaired users needing structured, accurate summaries.

ASPECT	Shortcomings of Existing Systems	Proposed Solution
Accessibility	Tools mostly cover translation; lack interactive summarization/Q&A for visually impaired Tamil users.	Tamil smart reader with real-time voice-based summarization, topic extraction, and Q&A.
Deployment	Cloud-dependent, resource-heavy unsuitable for mobile/low-connectivity.	Offline, on-device Tamil LLM optimized for lightweight mobile performance(gemma-3-270m).
Dataset Availability	No curated Tamil summarization/Q&A datasets shallow, fragmented outputs.	Build domain-specific Tamil summarization + QA dataset (crowdsourced + expert validation) with tamil-alpaca-orca
Performance	High latency, unsuitable for real-time reading or query response.	Optimized lightweight model ensuring fast response on-device.
User Experience	Limited interactivity, no voice-driven features for the visually impaired.	Voice-driven interactive reader enabling natural Q&A, structured summaries, and private use.

On-Device Tamil LLM

Key Domains of Specialization-based Knowledge Utilized

- **Natural Language Processing (NLP):**
  Utilized Tamil-specific LLM models for **topic extraction, summarization, and interactive Q&A** in Tamil. These models address key challenges in low-resource languages, improving accessibility for visually impaired users.
- **Machine Learning & Deep Learning:**
  Transformer-based architectures (e.g., **Gemma, TinyLLaMA**) are leveraged for **fine-tuning** and **model optimization**, ensuring efficient on-device deployment.
- **Mobile Computing:**
  Integration with **Android/iOS platforms** ensures compatibility with **smartphone accessibility tools** like **TalkBack**and **VoiceOver**, enhancing accessibility for visually impaired users.
- **Assistive Technology Design:**
  Incorporating **voice input/output** for hands-free interaction and **emotion-aware TTS** for more natural and engaging reading experiences.

Latest Technologies in the Domain Utilized

- **Transformer-based NLP Models:**
  Gemma-270M, TinyLLaMA, TAMIL-LLaMA models for on-device Tamil LLM deployment, focusing on **summarization** and **question answering**.
- **Model Optimization Techniques:**
  LoRA/QLoRA fine-tuning techniques to optimize large models for **mobile hardware** without relying on cloud infrastructure.
- **Speech-to-Text (STT) & Text-to-Speech (TTS):**
  Emotion-aware TTS for natural-sounding speech and **STT APIs** for **voice command inputs**, improving user interactivity.
- **Edge Computing:**
  On-device computing ensures **offline functionality**, with **low-latency** and **privacy preservation**.

On-Device Tamil LLM

How to Evaluate/Prove the Success

Accuracy Metrics:

- Summarization Quality: Measured using ROUGE score and BLEU score to ensure concise and relevant summaries.
- QA Performance: Evaluated using Exact Match and F1 score to validate the effectiveness of question answering.

Latency and Efficiency:

- Inference Latency: Should be ≤ 2 seconds for real-time summarization and Q&A.
- Memory Footprint and Power Usage: Ensure that the system runs efficiently on mobile hardware, optimizing for battery consumption and processing power.

Usability Testing:

- Conduct user studies with visually impaired participants to gather feedback on clarity, usability, and accessibility.
- Iterative User Feedback: Based on feedback, the design and performance of the application will be refined.

Data Availability and Ethical Clearance

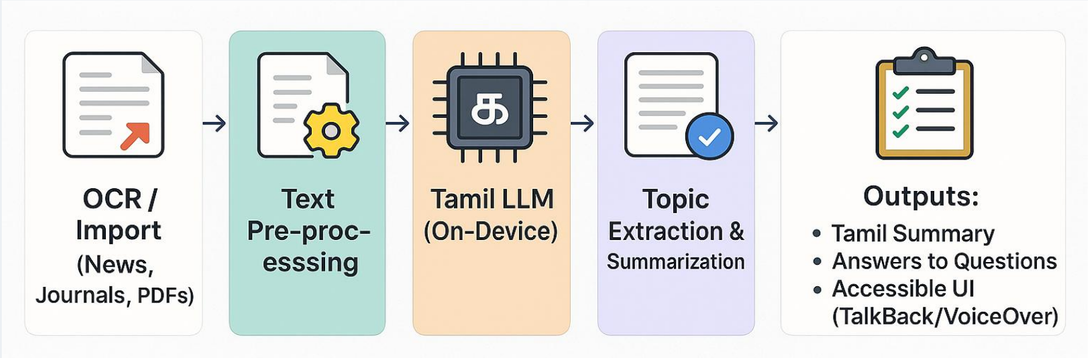
Data Availability:

- Tamil Corpora for Summarization & QA: Use of existing datasets like CulturaX-ta, Alpaca-Orca, and Wikipedia dumps, augmented with web scraping for additional Tamil text from newspapers, journals, and educational resources.
- Ethical Considerations in Data Collection:
  - Informed consent for any crowdsourced annotations or validation.
  - Ensure that all scraped data is publicly available and follows ethical guidelines to respect privacy.

Ethical Clearance:

- Human-Centered Design: Ethical approval will be sought for conducting user studies with visually impaired individuals to ensure that their participation is voluntary and informed.
- Data Privacy: Offline processing ensures that user data is not stored or transmitted, maintaining user privacy throughout the interaction.

On-Device Tamil LLM



Real-World Usage Scenarios

- For Visually Impaired Students:
  Point the camera at a text to extract and summarize it in Tamil; ask for clarifications on specific terms.
- For Elderly or Rural Users:
  Works offline, providing access to Tamil news, books, and educational content without internet.
- Daily News or Journal Access:
  Allows visually impaired users to listen to news and ask specific questions about the content.
- Integration with Accessibility Tools:
  Seamlessly integrates with TalkBack (Android) and VoiceOver (iOS) for improved accessibility.

System Overview of the Individual Component

Smart Reader Application Architecture:

- The system is a mobile-based application designed for visually impaired users, integrating OCR (Optical Character Recognition)for text extraction, Tamil LLM for summarization and topic extraction, and emotion-aware TTS for natural speech synthesis.

Key Functional Components:

- Text Capture Module: Users capture images of text using their smartphone camera.
- OCR Processing: Extracts text from images (optimized for Sinhala and Tamil).
- Tamil LLM: Processes the text for topic extraction, summarization, and interactive Q&A.
- TTS Module: Converts the summarized text or answers into speech with emotional cues for a natural listening experience.
- Voice Command Interface: Users can ask questions or request summaries via voice input, making it hands-free andfully interactive.

Offline Deployment:

- The entire system works offline, ensuring users in low-connectivity areas can still access the content.

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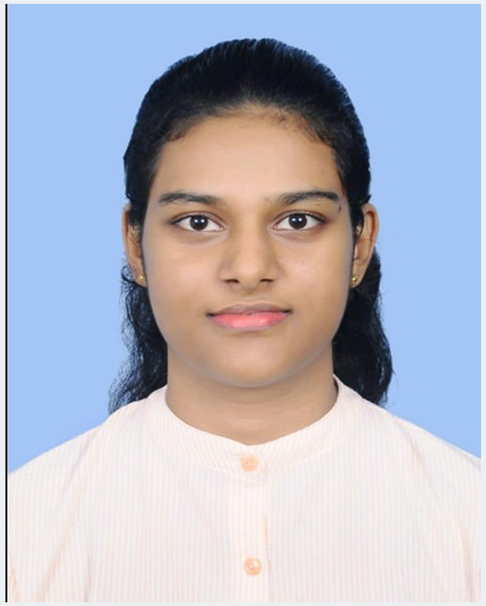
Data Science



Emotional TTS

•**Proven Gap:** Sinhala TTS systems are monotone and robotic due to the lack of emotional datasets and optimized low-resource models.

Solution

• **Creative Solution:** Develop an emotion-aware Sinhala TTS smart reader app using a new emotional speech dataset and optimized deep learning models for natural, expressive real-time speech.

Knowledge Gap / Problem Definition: Sinhala lacks emotional datasets, phonetic tools, and affordable assistive readers, leading to flat voices and limited accessibility for visually impaired users.

ASPECT	Shortcomings of Existing Systems	Proposed Solution
Accessibility	Mostly English-based, costly, not tailored for Sinhala.	Low-cost Sinhala/Tamil mobile reader with basic emotion-aware TTS.
Emotional Speech Data	Lack of Sinhala emotional datasets available.	Start building a small but reliable Sinhala emotional speech dataset (expert + limited crowd validation).
Phonetic Accuracy	Weak phonetic tools → frequent mispronunciations	Apply G2P rules and iterative corrections for better accuracy.
Performance	High latency → not practical for smooth real-time reading	Lightweight models optimized for faster response on mobile devices.
Voice Quality	Flat, robotic, monotone voices with no emotion	Add basic emotional tones (happy, sad, neutral) for more natural speech.

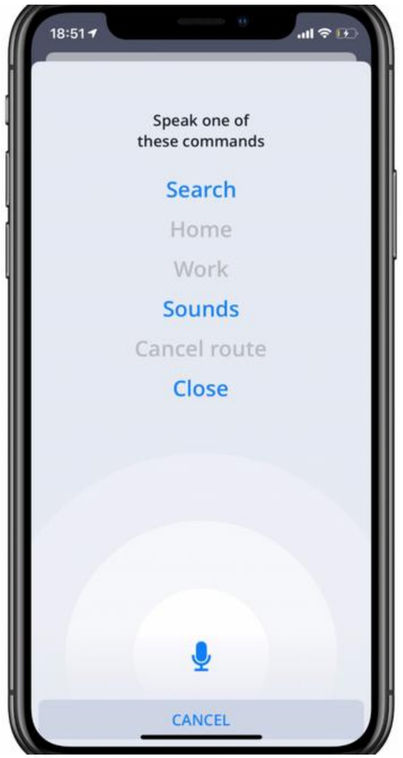
Emotional TTS

Key Domains of Specialization

•Data Science & AI → Speech processing, NLP, and emotion recognition.
•Deep Learning & TTS Models → Tacotron 2, FastSpeech 2, VAENAR-TTS, RAD-TTS for expressive speech synthesis.
•Accessibility & HCI(Human-Computer Interaction) → Inclusive assistive technology for visually impaired users.

Latest Technologies Utilized

- Transformer Models → Sinhala-BERT, XLM-R for emotion & sentiment detection.
- Non-Autoregressive TTS → VAENAR-TTS and FastSpeech 2 for real-time performance.
- Global Style Tokens (GSTs) → Emotion and style transfer in speech.
- Web Scraping Tools → yt-dlp, Scrapy, Selenium.
- Cloud Platforms → Google Colab Pro, RunPod for training & optimization.
- Natural Vocoders → HiFi-GAN, Wave-Glow.



Emotional TTS

Validation & Data/Ethics

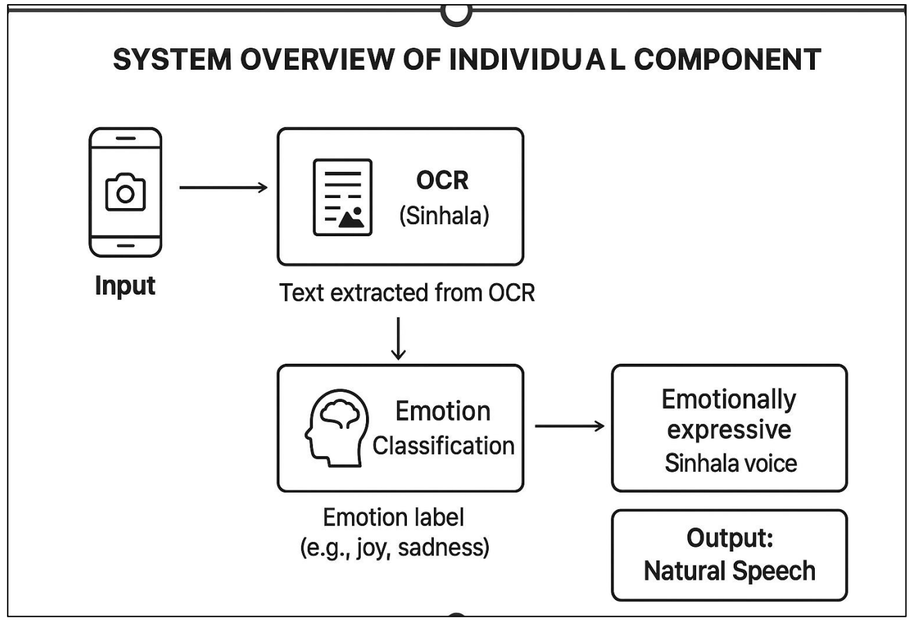
How to Evaluate/Prove Success

•Objective Metrics: Word Error Rate (WER), Mel-Cepstral Distortion (MCD), Inference Speed (real-time)
•Subjective Metrics: Mean Opinion Score (MOS) from listeners, user studies with visually impaired participants
•Expected Outcome: ≥ 75% inter-annotator agreement, MOS ≥ 4.0 for naturalness and expressiveness, and RFT<0.5 for real time speech generation.

Data Availability & Ethical Clearance

- Dataset Collection: Volunteer-based Sinhala emotional speech, ethical web-scraping (YouTube, podcasts) with consent
- Ethical Considerations: Full **informed consent** from volunteers and annotators, **Anonymity and privacy** preserved in all data releases and Dataset used strictly for **research and accessibility purposes.**

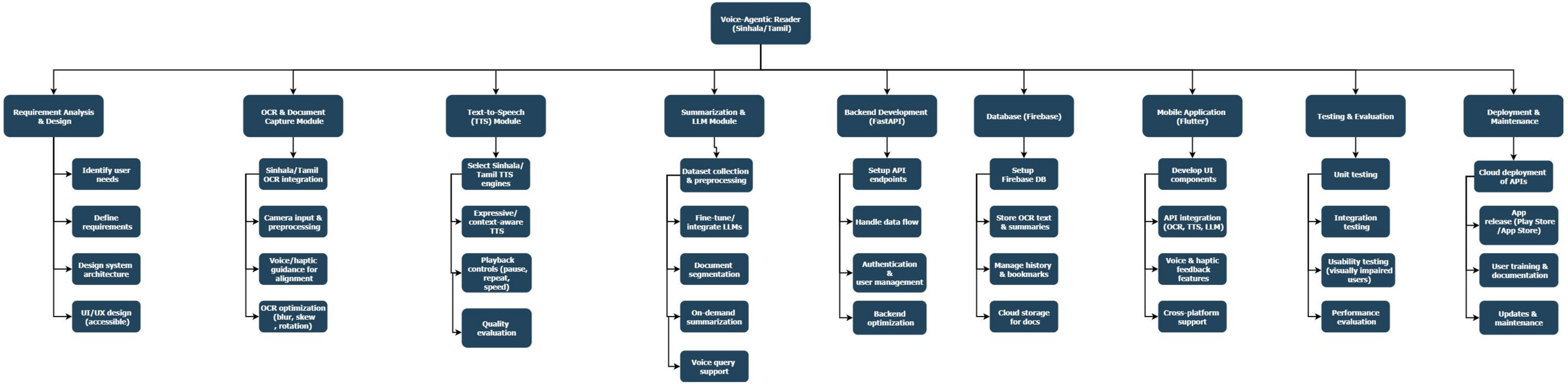
Emotional TTS



- A Sinhala visual impaired person is reading a digital storybook.
- The app scans the page and **extracts the text** (OCR).
- The system **detects emotions** in the sentences (e.g., "ඌ නා ඉ නා ම සතු වූ වූ ණ " → 😊joy)
- The **TTS reads aloud** the story with matching emotions happy parts in a cheerful tone, sad parts in a softer tone.
- The student **hears the story naturally**, making it more engaging and easier to understand.

Functional requirements	Non functional requirements
• Accept Sinhala text input	• Simple user interface
• Convert text to phonemes	• Fast response (low latency)
• Generate Sinhala speech output	• High accuracy & naturalness
• Support multiple voices/emotions	• User-friendly and accessible
• allow pitch/speed/volume adjustments;	• Scalable & reliable
	• Secure handling of user data

WORK BREAKDOWN STRUCTURE



SUPPORTIVE INFORMATION BUDGET

- Development Cost
- Server & Hosting Cost
- Marketing Budget
- Purchasing necessary Softwares
- Maintenance Budget
- Legal & Administrative Expenses

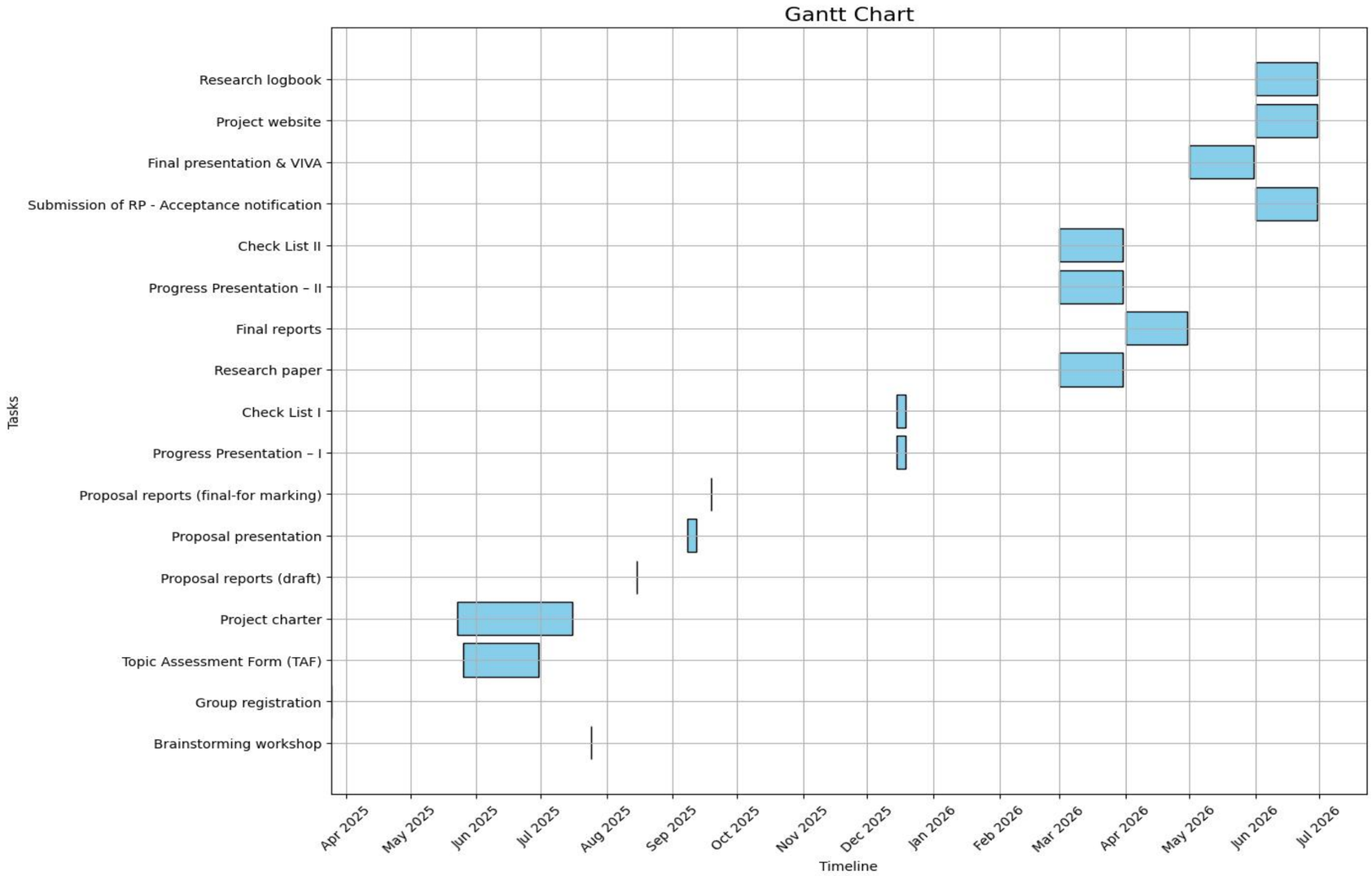
Commercialization

- **Identifying the Target Audience: In this case, the target audience would be visually Impaired Individuals who could benefit from the features of the Mobile Application.**

- **Revenue Generation: Commercialization also involves generating revenue from the mobile application. This could involve charging for the application itself, offering premium features for a fee.**

- **Promotions: The target audience should be reached through targeted campaigns on social media and other channels. Collaboration with organizations that work with visually impaired individuals, such as libraries or schools, can also be an effective way to reach the target audience.**

GANTT CHART


Gantt Chart
Research logbook
Project website
Final presentation & VIVA
Submission of RP - Acceptance notification
Check List II
Progress Presentation – II
Final reports
Research paper
Check List I
Progress Presentation – I
Proposal reports (final-for marking)
Proposal presentation
Proposal reports (draft)
Project charter
Topic Assessment Form (TAF)
Group registration
Brainstorming workshop
Tasks
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Timeline

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Thank You

Any Questions